

TURKMENABAT, Turkmenistan

WMO: 386870

Lat: 39.083N

Lon: 63.600E

Elev: 623

StdP: 14.37

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	19.0	4.3	7.0	18.2	10.0	9.4	25.3	21.3	54.5	18.8	48.6	5.6	0	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	103.4	69.1	100.7	68.4	98.4	67.8	72.8	95.8	71.1	94.3	69.7	93.0	9.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	95.2	83.5	63.0	88.1	81.9	61.0	82.1	81.1	36.9	95.7	35.3	94.5	34.1	93.1	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.4	14.4	DB	9.3	107.7	7.5	2.5	3.9	109.5	-0.5	111.0	-4.6	112.4	-10.1	114.2	(4)
(5)			WB	8.4	77.1	7.4	2.5	3.1	78.9	-1.2	80.4	-5.4	81.8	-10.8	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	37.6	41.3	52.7	64.2	74.7	82.2	84.6	81.4	71.4	59.9	47.7	39.1		
	DBStd	18.49	9.47	10.19	9.12	8.56	5.95	4.72	4.58	4.88	6.06	7.90	9.19	8.75		
	HDD50	1259	394	270	77	6	0	0	0	0	0	15	144	353		
	HDD65	3561	850	664	390	116	6	0	0	0	0	194	521	804		
	CDD50	5463	9	26	162	432	767	965	1074	974	642	322	76	14		
	CDD65	2290	0	0	10	92	308	515	609	509	208	36	3	0		
	CDH74	30710	0	1	144	1045	3719	7267	8753	6738	2478	519	46	0		
	CDH80	17206	0	0	31	385	1805	4320	5424	3978	1119	137	7	0		
(14) Wind	WSAvg	6.8	6.5	7.3	7.7	7.3	7.0	7.5	7.8	7.3	6.5	5.5	5.5	6.2		
(15) Precipitation	PrecAvg	5.0	0.7	0.8	1.1	0.8	0.4	0.0	0.0	0.0	0.0	0.1	0.5	0.6		
	PrecMax	8.6	2.5	2.3	2.5	3.6	2.2	0.4	0.1	0.1	0.9	0.7	1.7	2.6		
	PrecMin	1.9	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.5	0.6	0.6	0.6	0.7	0.5	0.1	0.0	0.0	0.1	0.2	0.4	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.8	71.2	84.1	95.0	100.0	104.8	108.3	104.7	97.3	89.6	79.7	67.7		
		MCWB	49.5	53.1	59.5	63.7	67.4	69.6	69.6	69.0	66.7	63.5	58.0	49.8		
	2%	DB	60.4	65.6	78.2	88.8	96.2	101.7	104.4	101.7	93.8	83.7	72.7	61.4		
		MCWB	47.3	50.6	57.5	62.7	65.7	68.3	69.9	68.6	65.3	60.5	55.4	47.9		
	5%	DB	55.7	60.9	73.1	84.4	93.1	99.3	101.4	98.8	90.6	79.8	67.0	56.9		
		MCWB	45.8	48.7	55.5	60.9	64.6	67.4	68.8	68.3	64.1	59.0	52.8	46.5		
	10%	DB	50.7	56.3	68.2	80.3	89.7	96.8	98.5	96.2	86.7	75.8	62.5	52.1		
		MCWB	43.6	46.9	53.7	59.3	63.7	66.8	68.4	66.9	62.5	56.9	50.7	44.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	55.1	61.7	67.7	71.7	73.6	76.3	73.8	70.2	65.5	59.9	52.1		
		MCDB	61.4	66.1	77.8	84.8	91.8	97.8	99.2	95.8	91.6	84.5	74.4	62.1		
	2%	WB	48.7	52.4	58.8	65.0	68.5	71.5	73.2	71.7	67.4	62.1	56.8	49.6		
		MCDB	58.1	62.8	73.3	83.4	89.5	94.7	96.5	94.7	88.9	80.0	70.1	59.1		
	5%	WB	46.3	50.0	56.7	62.8	66.8	69.7	71.4	70.0	65.3	60.1	54.2	47.2		
		MCDB	54.5	58.8	70.3	79.6	87.4	93.6	94.9	93.0	86.5	76.5	65.0	55.5		
	10%	WB	43.9	47.5	54.5	61.0	65.2	68.2	69.9	68.5	63.4	58.2	51.8	44.9		
		MCDB	50.2	55.4	66.7	76.6	84.9	92.0	93.7	91.9	84.3	73.5	61.0	51.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	18.4	20.6	22.9	25.1	26.3	25.5	27.3	28.1	26.3	21.0	16.3		
		MCDBR	23.7	25.1	27.8	29.1	29.7	29.8	28.7	30.4	32.0	30.8	27.1	23.2		
		MCWBR	13.2	12.9	11.7	10.1	9.9	10.2	9.9	11.2	12.7	13.3	12.7	12.9		
	5% WB	MCDBR	21.7	22.2	25.3	25.4	26.1	27.8	26.9	28.8	30.5	28.8	24.6	21.2		
		MCWBR	13.0	12.3	11.3	9.9	10.3	10.5	10.3	11.6	12.8	13.2	12.4	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.383	0.427	0.500	0.531	0.529	0.531	0.500	0.458	0.437	0.474	0.444	0.390		
	taud		2.112	2.001	1.851	1.816	1.847	1.844	1.922	2.018	2.032	1.912	1.969	2.079		
	Ebn at Noon		238	243	237	240	244	243	249	257	253	223	211	223		
	Edh at Noon		36	46	60	66	65	65	60	53	49	50	42	35		
(44) All-Sky Solar Radiation	RadAvg		665	974	1342	1787	2195	2491	2465	2262	1854	1289	821	597		
	RadStd		61	98	102	139	89	73	51	48	47	71	75	63		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	+1.40	-0.97	-1.15	N/A	N/A	+234	+230
(47) Regional (8 neighbors)		N/A	N/A	N/A	+0.56	-1.22	-1.66	N/A	N/A	N/A	+90

Nomenclature: See separate page